“Data Cleaning, Analysis & Visualization with Excel”

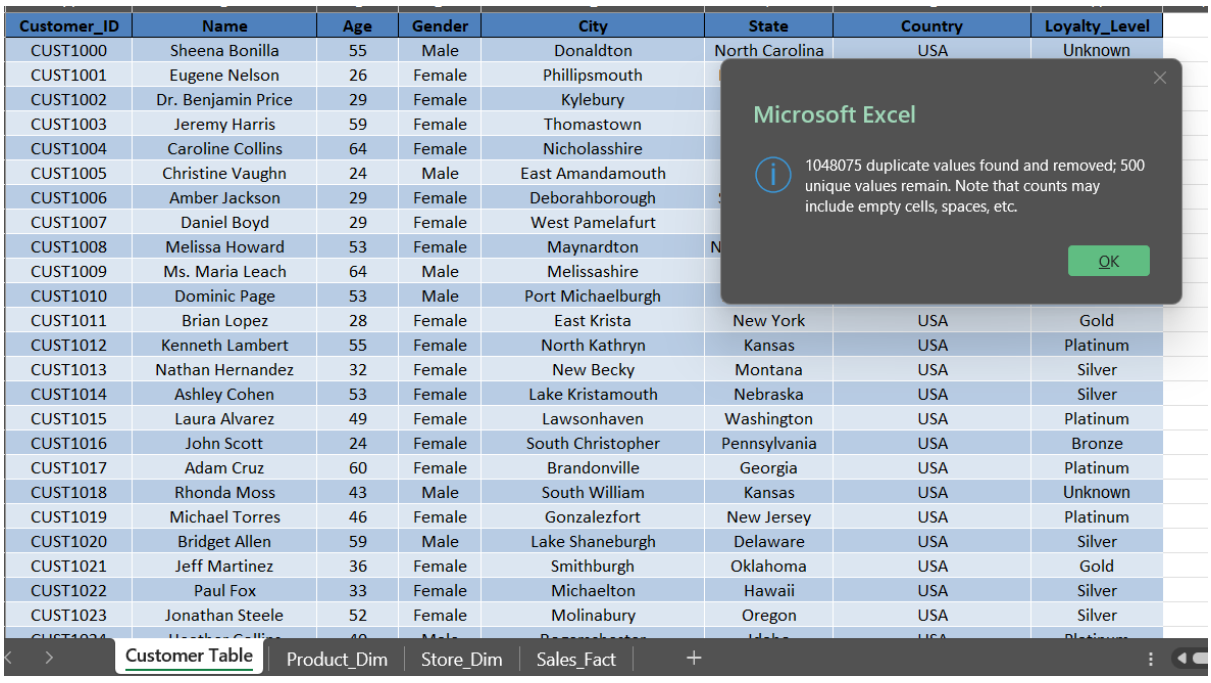
1. Data entry, import, and basic organization using tables

Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	Unknown
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	Unknown
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA	Silver

Customer Table | Product_Dim | Store_Dim | Sales_Fact

2. Data cleaning: find/remove duplicates, handle missing/inconsistent data, correct formatting errors, and validate data

    1. Find and remove duplicates



2. Handle missing/inconsistent data, correct formatting errors, and validate data

Calculated_Amount in Sales_Fact

=F2*G2*(1-H2)

Calculate Quantity

=J2/(G2*(1-H2))

calculate Unit Price

=J2/(F2*(1-H2))

calculate Discount

=1-(J28/(F28*G28))

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Calculated_Amount	Calculate Quantity	calculate Unit Price	calculate Discount
SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	₹218.04	9.00%	Credit Card	198.42	198.42	1	₹218.04	9.00%
SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	₹104.47	27.00%	PayPal	228.79	228.79	3	₹104.47	27.00%
SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	₹914.05	25.00%	Credit Card	2742.15	2742.15	4	₹914.05	25.00%
SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	₹921.51	16.00%	COD	774.07	774.07	1	₹921.51	16.00%
SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	₹745.30	18.00%	Credit Card	1833.44	1833.44	3	₹745.30	18.00%
SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	₹147.01	19.00%	COD	357.23	357.23	3	₹147.01	19.00%
SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	₹785.31	15.00%	PayPal	1335.03	1335.03	2	₹785.31	15.00%
SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	₹342.39	23.00%	PayPal	1054.56	1054.56	4	₹342.39	23.00%
SALE8	04-05-2025	CUST1187	PROD99	STORE5	**3**	₹448.25	7.00%	PayPal	1250.62		3	₹448.25	7.00%
SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	₹873.17	23.00%	PayPal	1344.68	1344.68	2	₹873.17	23.00%
SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	₹278.63	7.00%	Credit Card	259.13	259.13	1	₹278.63	7.00%
SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	₹286.92	17.00%	Credit Card	952.57	952.57	4	₹286.92	17.00%
SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	₹644.58	1.00%	Credit Card	2552.54	2552.54	4	₹644.58	1.00%
SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	₹316.90	30.00%	PayPal	443.66	443.66	2	₹316.90	30.00%
SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	₹403.47	29.00%	PayPal	1145.85	1145.85	4	₹403.47	29.00%
SALE15	10-12-2024	CUST1435	PROD36	STORE9	**3**	₹871.34	4.00%	COD	2509.46		3	₹871.34	4.00%
SALE16	13-09-2025	CUST1276	PROD65	STORE19	2	₹451.98	25.00%	Credit Card	677.97	677.97	2	₹451.98	25.00%
SALE17	11-12-2024	CUST1079	PROD27	STORE4	2	₹502.69	23.00%	Credit Card	774.14	774.14	2	₹502.69	23.00%
SALE18	13-08-2024	CUST1094	PROD39	STORE4	2	₹559.23	25.00%	COD	838.85	838.85	2	₹559.23	25.00%
SALE19	18-03-2024	CUST1239	PROD60	STORE13	4	**₹464.18**	4.00%	COD	1782.45			₹464.18	4.00%
SALE20	05-07-2025	CUST1144	PROD72	STORE17	3	₹986.04	27.00%	COD	2159.43	2159.43	3	₹986.04	27.00%
SALE21	04-06-2024	CUST1396	PROD43	STORE18	1	₹589.56	22.00%	Credit Card	459.86	459.86	1	₹589.56	22.00%
SALE22	12-04-2024	CUST1476	PROD69	STORE10	3	₹979.42	3.00%	PayPal	2850.11	2850.11	3	₹979.42	3.00%
SALE23	23-06-2024	CUST1368	PROD12	STORE8	1	₹719.28	8.00%	Credit Card	661.74	661.74	1	₹719.28	8.00%

Calculate total sales

=SUM(J2:J2001)

Calculate average sales

=AVERAGE(J2:J2001)

Count transactions by payment type

=COUNTIF(I2:I2001,"PayPal")

Categorize sales as high or low value

=IF(J2>=1000,"High Value","Low Value")

Extract month and year from a date

=TEXT(B2,"mmm-yyyy")

Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Total sales	Avg sales	Count transby pyt	High or low value	Extract month and year
07-07-2024	CUST1409	PROD97	STORE3	1	₹ 218.04	9.00%	Credit Card	198.42	2174394.70	1087.20	695.00	Low Value	Jul-2024
25-12-2023	CUST1467	PROD74	STORE17	3	₹ 104.47	27.00%	PayPal	228.79	2174196.28	1087.64	695.00	Low Value	Dec-2023
03-06-2024	CUST1056	PROD49	STORE8	4	₹ 914.05	25.00%	Credit Card	2742.15	2173967.49	1088.07	694.00	High Value	Jun-2024
29-11-2023	CUST1346	PROD76	STORE8	1	₹ 921.51	16.00%	COD	774.07	2171225.34	1087.24	694.00	Low Value	Nov-2023
06-01-2024	CUST1493	PROD59	STORE5	3	₹ 745.30	18.00%	Credit Card	1833.44	2170451.27	1087.40	694.00	High Value	Jan-2024
26-07-2024	CUST1116	PROD15	STORE7	3	₹ 147.01	19.00%	COD	357.23	2168617.84	1087.03	694.00	Low Value	Jul-2024
12-07-2025	CUST1016	PROD56	STORE4	2	₹ 785.31	15.00%	PayPal	1335.03	2168260.60	1087.39	694.00	High Value	Jul-2025
18-11-2023	CUST1152	PROD70	STORE8	4	₹ 342.39	23.00%	PayPal	1054.56	2166925.57	1087.27	693.00	High Value	Nov-2023
04-05-2025	CUST1187	PROD99	STORE5	**3**	₹ 448.25	7.00%	PayPal	1250.62	2165871.01	1087.28	692.00	High Value	May-2025
31-07-2025	CUST1464	PROD33	STORE0	2	₹ 873.17	23.00%	PayPal	1344.68	2164620.40	1087.20	691.00	High Value	Jul-2025
13-02-2024	CUST1065	PROD24	STORE14	1	₹ 278.63	7.00%	Credit Card	259.13	2163275.71	1087.07	690.00	Low Value	Feb-2024
24-05-2025	CUST1320	PROD39	STORE1	4	₹ 286.92	17.00%	Credit Card	952.57	2163016.59	1087.49	690.00	Low Value	May-2025
24-08-2024	CUST1237	PROD88	STORE4	4	₹ 644.58	1.00%	Credit Card	2552.54	2162064.01	1087.56	690.00	High Value	Aug-2024
06-08-2025	CUST1443	PROD34	STORE16	2	₹ 316.90	30.00%	PayPal	443.66	2159511.48	1086.82	690.00	Low Value	Aug-2025
21-06-2025	CUST1348	PROD24	STORE15	4	₹ 403.47	29.00%	PayPal	1145.85	2159067.82	1087.14	689.00	High Value	Jun-2025
10-12-2024	CUST1435	PROD36	STORE9	**3**	₹ 871.34	4.00%	COD	2509.46	2157921.96	1087.11	688.00	High Value	Dec-2024
13-09-2025	CUST1276	PROD65	STORE19	2	₹ 451.98	25.00%	Credit Card	677.97	2155412.50	1086.40	688.00	Low Value	Sep-2025
11-12-2024	CUST1079	PROD27	STORE4	2	₹ 502.69	23.00%	Credit Card	774.14	2154734.53	1086.60	688.00	Low Value	Dec-2024
13-08-2024	CUST1094	PROD39	STORE4	2	₹ 559.23	25.00%	COD	838.85	2153960.39	1086.76	688.00	Low Value	Aug-2024
18-03-2024	CUST1239	PROD60	STORE13	4	**₹ 464.18**	4.00%	COD	1782.45	2153121.55	1086.89	688.00	High Value	Mar-2024
05-07-2025	CUST1144	PROD72	STORE17	3	₹ 986.04	27.00%	COD	2159.43	2151339.09	1086.53	688.00	High Value	Jul-2025
04-06-2024	CUST1396	PROD43	STORE18	1	₹ 589.56	22.00%	Credit Card	459.86	2149179.67	1085.99	688.00	Low Value	Jun-2024
12-04-2024	CUST1476	PROD69	STORE10	3	₹ 979.42	3.00%	PayPal	2850.11	2148719.81	1086.31	688.00	High Value	Apr-2024
23-06-2024	CUST1368	PROD12	STORE8	1	₹ 719.28	8.00%	Credit Card	661.74	2145869.70	1085.42	687.00	Low Value	Jun-2024

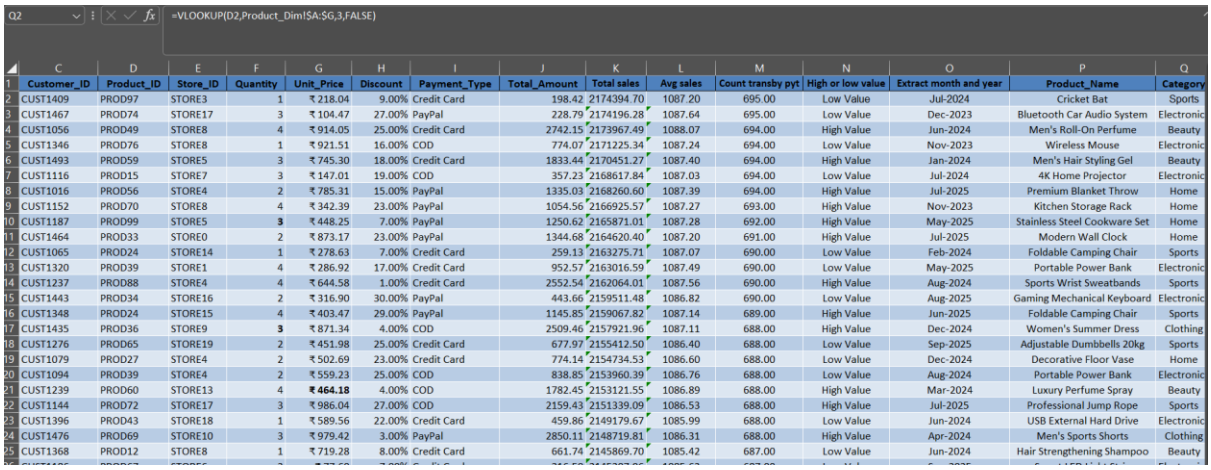
Customer Table | Product_Dim | Store_Dim | Sales_Fact

XLOOKUP for Sales_Fact and Product_Dim (Product_Name)



Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Total sales	Avg sales	Count transby pyt	High or low value	Extract month and year	Product_Name
CUST1409	PROD97	STORE3	1	₹ 218.04	9.00%	Credit Card	198.42	2174394.70	1087.20	695.00	Low Value	Jul-2024	=XLOOKUP([@[Product_ID]],
CUST1467	PROD74	STORE17	3	₹ 104.47	27.00%	PayPal	228.79	2174196.28	1087.64	695.00	Low Value	Dec-2023	Bluetooth Car Audio System
CUST1056	PROD49	STORE8	4	₹ 914.05	25.00%	Credit Card	2742.15	2173967.49	1088.07	694.00	High Value	Jun-2024	Men's Roll-On Perfume
CUST1346	PROD76	STORE8	1	₹ 921.51	16.00%	COD	774.07	2171225.34	1087.24	694.00	Low Value	Nov-2023	Wireless Mouse
CUST1493	PROD59	STORE5	3	₹ 745.30	18.00%	Credit Card	1833.44	2170451.27	1087.40	694.00	High Value	Jan-2024	Men's Hair Styling Gel
CUST1116	PROD15	STORE7	3	₹ 147.01	19.00%	COD	357.23	2168617.84	1087.03	694.00	Low Value	Jul-2024	4K Home Projector
CUST1016	PROD56	STORE4	2	₹ 785.31	15.00%	PayPal	1335.03	2168260.60	1087.39	694.00	High Value	Jul-2025	Premium Blanket Throw
CUST1152	PROD70	STORE8	4	₹ 342.39	23.00%	PayPal	1054.56	2166925.57	1087.27	693.00	High Value	Nov-2023	Kitchen Storage Rack
CUST1187	PROD99	STORE5	3	₹ 448.25	7.00%	PayPal	1250.62	2165871.01	1087.28	692.00	High Value	May-2025	Stainless Steel Cookware Set
CUST1464	PROD33	STORE0	2	₹ 873.17	23.00%	PayPal	1344.68	2164620.40	1087.20	691.00	High Value	Jul-2025	Modern Wall Clock
CUST1065	PROD24	STORE14	1	₹ 278.63	7.00%	Credit Card	259.13	2163275.71	1087.07	690.00	Low Value	Feb-2024	Foldable Camping Chair
CUST1320	PROD39	STORE1	4	₹ 286.92	17.00%	Credit Card	952.57	2163016.59	1087.49	690.00	Low Value	May-2025	Portable Power Bank
CUST1237	PROD88	STORE4	4	₹ 644.58	1.00%	Credit Card	2552.54	2162064.01	1087.56	690.00	High Value	Aug-2024	Sports Wrist Sweatbands
CUST1443	PROD34	STORE16	2	₹ 316.90	30.00%	PayPal	443.66	2159511.48	1086.82	690.00	Low Value	Aug-2025	Gaming Mechanical Keyboard
CUST1348	PROD24	STORE15	4	₹ 403.47	29.00%	PayPal	1145.85	2159067.82	1087.14	689.00	High Value	Jun-2025	Foldable Camping Chair
CUST1435	PROD36	STORE9	3	₹ 871.34	4.00%	COD	2509.46	2157921.96	1087.11	688.00	High Value	Dec-2024	Women's Summer Dress
CUST1276	PROD65	STORE19	2	₹ 451.98	25.00%	Credit Card	677.97	2155412.50	1086.40	688.00	Low Value	Sep-2025	Adjustable Dumbbells 20kg
CUST1079	PROD27	STORE4	2	₹ 502.69	23.00%	Credit Card	774.14	2154734.53	1086.60	688.00	Low Value	Dec-2024	Decorative Floor Vase
CUST1094	PROD39	STORE4	2	₹ 559.23	25.00%	COD	838.85	2153960.39	1086.76	688.00	Low Value	Aug-2024	Portable Power Bank
CUST1239	PROD60	STORE13	4	₹ 464.18	4.00%	COD	1782.45	2153121.55	1086.89	688.00	High Value	Mar-2024	Luxury Perfume Spray
CUST1144	PROD72	STORE17	3	₹ 986.04	27.00%	COD	2159.43	2151339.09	1086.53	688.00	High Value	Jul-2025	Professional Jump Rope
CUST1396	PROD43	STORE18	1	₹ 589.56	22.00%	Credit Card	459.86	2149179.67	1085.99	688.00	Low Value	Jun-2024	USB External Hard Drive
CUST1476	PROD69	STORE10	3	₹ 979.42	3.00%	PayPal	2850.11	2148719.81	1086.31	688.00	High Value	Apr-2024	Men's Sports Shorts
CUST1368	PROD12	STORE8	1	₹ 719.28	8.00%	Credit Card	661.74	2145869.70	1085.42	687.00	Low Value	Jun-2024	Hair Strengthening Shampoo

VLOOKUP for Sales_Fact and Product_Dim (Category)



Q2 =VLOOKUP(D2,Product_Dim!$A:$G,3,FALSE)

Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Total sales	Avg sales	Count transby pyt	High or low value	Extract month and year	Product_Name	Category
CUST1409	PROD97	STORE3	1	₹218.04	9.00%	Credit Card	198.42	2174394.70	1087.20	695.00	Low Value	Jul-2024	Cricket Bat	Sports
CUST1467	PROD74	STORE17	3	₹104.47	27.00%	PayPal	228.79	2174196.28	1087.64	695.00	Low Value	Dec-2023	Bluetooth Car Audio System	Electronic
CUST1056	PROD49	STORE8	4	₹914.05	25.00%	Credit Card	2742.15	2173967.49	1088.07	694.00	High Value	Jun-2024	Men's Roll-On Perfume	Beauty
CUST1346	PROD76	STORE8	1	₹921.51	16.00%	COD	774.07	2171225.34	1087.24	694.00	Low Value	Nov-2023	Wireless Mouse	Electronic
CUST1493	PROD59	STORE5	3	₹745.30	18.00%	Credit Card	1833.44	2170451.27	1087.40	694.00	High Value	Jan-2024	Men's Hair Styling Gel	Beauty
CUST1116	PROD15	STORE7	3	₹147.01	19.00%	COD	357.23	2168617.84	1087.03	694.00	Low Value	Jul-2024	4K Home Projector	Electronic
CUST1016	PROD56	STORE4	2	₹785.31	15.00%	PayPal	1335.03	2168260.60	1087.39	694.00	High Value	Jul-2025	Premium Blanket Throw	Home
CUST1152	PROD70	STORE8	4	₹342.39	23.00%	PayPal	1054.56	2166925.57	1087.27	693.00	High Value	Nov-2023	Kitchen Storage Rack	Home
CUST1187	PROD99	STORE5	**3**	₹448.25	7.00%	PayPal	1250.62	2165871.01	1087.28	692.00	High Value	May-2025	Stainless Steel Cookware Set	Home
CUST1464	PROD33	STORE0	2	₹873.17	23.00%	PayPal	1344.68	2164620.40	1087.20	691.00	High Value	Jul-2025	Modern Wall Clock	Home
CUST1065	PROD24	STORE14	1	₹278.63	7.00%	Credit Card	259.13	2163275.71	1087.07	690.00	Low Value	Feb-2024	Foldable Camping Chair	Sports
CUST1320	PROD39	STORE1	4	₹286.92	17.00%	Credit Card	952.57	2163016.59	1087.49	690.00	Low Value	May-2025	Portable Power Bank	Electronic
CUST1237	PROD88	STORE4	4	₹644.58	1.00%	Credit Card	2552.54	2162064.01	1087.56	690.00	High Value	Aug-2024	Sports Wrist Sweatbands	Sports
CUST1443	PROD34	STORE16	2	₹316.90	30.00%	PayPal	443.66	2159511.48	1086.82	690.00	Low Value	Aug-2025	Gaming Mechanical Keyboard	Electronic
CUST1348	PROD24	STORE15	4	₹403.47	29.00%	PayPal	1145.85	2159067.82	1087.14	689.00	High Value	Jun-2025	Foldable Camping Chair	Sports
CUST1435	PROD36	STORE9	**3**	₹871.34	4.00%	COD	2509.46	2157921.96	1087.11	688.00	High Value	Dec-2024	Women's Summer Dress	Clothing
CUST1276	PROD65	STORE19	2	₹451.98	25.00%	Credit Card	677.97	2155412.50	1086.40	688.00	Low Value	Sep-2025	Adjustable Dumbbells 20kg	Sports
CUST1079	PROD27	STORE4	2	₹502.69	23.00%	Credit Card	774.14	2154734.53	1086.60	688.00	Low Value	Dec-2024	Decorative Floor Vase	Home
CUST1094	PROD39	STORE4	2	₹559.23	25.00%	COD	838.85	2153960.39	1086.76	688.00	Low Value	Aug-2024	Portable Power Bank	Electronic
CUST1239	PROD60	STORE13	4	**₹464.18**	4.00%	COD	1782.45	2153121.55	1086.89	688.00	High Value	Mar-2024	Luxury Perfume Spray	Beauty
CUST1144	PROD72	STORE17	3	₹986.04	27.00%	COD	2159.43	2151339.09	1086.53	688.00	High Value	Jul-2025	Professional Jump Rope	Sports
CUST1396	PROD43	STORE18	1	₹589.56	22.00%	Credit Card	459.86	2149179.67	1085.99	688.00	Low Value	Jun-2024	USB External Hard Drive	Electronic
CUST1476	PROD69	STORE10	3	₹979.42	3.00%	PayPal	2850.11	2148719.81	1086.31	688.00	High Value	Apr-2024	Men's Sports Shorts	Clothing
CUST1368	PROD12	STORE8	1	₹719.28	8.00%	Credit Card	661.74	2145869.70	1085.42	687.00	Low Value	Jun-2024	Hair Strengthening Shampoo	Beauty

PIVOT



Avg sales	Count transby p	High or low val	Extract month and ye	Product_Name	Categor
1087.20	695.00	Low Value	Jul-2024	Cricket Bat	Sports
1087.64	695.00	Low Value	Dec-2023	Bluetooth Car Audio System	Electronics
1088.07	694.00	High Value	Jun-2024	Men's Roll-On Perfume	Beauty
1087.24	694.00	Low Value	Nov-2023	Wireless Mouse	Electronics
1087.40	694.00	High Value	Jan-2024	Men's Hair Styling Gel	Beauty
1087.03	694.00	Low Value	Jul-2024	4K Home Projector	Electronics
1087.39	694.00	High Value	Jul-2025	Premium Blanket Throw	Home
1087.27	693.00	High Value	Nov-2023	Kitchen Storage Rack	Home
1087.28	692.00	High Value	May-2025	Stainless Steel Cookware Set	Home
1087.20	691.00	High Value	Jul-2025	Modern Wall Clock	Home
1087.07	690.00	Low Value	Feb-2024	Foldable Camping Chair	Sports
1087.49	690.00	Low Value	May-2025	Portable Power Bank	Electronics
1087.56	690.00	High Value	Aug-2024	Sports Wrist Sweatbands	Sports
1086.82	690.00	Low Value	Aug-2025	Gaming Mechanical Keyboard	Electronics
1087.14	689.00	High Value	Jun-2025	Foldable Camping Chair	Sports
1087.11	688.00	High Value	Dec-2024	Women's Summer Dress	Clothing
1086.40	688.00	Low Value	Sep-2025	Adjustable Dumbbells 20kg	Sports
1086.60	688.00	Low Value	Dec-2024	Decorative Floor Vase	Home
1086.76	688.00	Low Value	Aug-2024	Portable Power Bank	Electronics
1086.89	688.00	High Value	Mar-2024	Luxury Perfume Spray	Beauty
1086.53	688.00	High Value	Jul-2025	Professional Jump Rope	Sports
1085.99	688.00	Low Value	Jun-2024	USB External Hard Drive	Electronics
1086.31	688.00	High Value	Apr-2024	Men's Sports Shorts	Clothing
1085.42	687.00	Low Value	Jun-2024	Hair Strengthening Shampoo	Beauty

Row Labels	Sum of Total_Amount
Beauty	321561.3283
Clothing	379216.0438
Electronics	504311.348
Home	427295.8036
Sports	542010.1746
(blank)	
Grand Total	**2174394.698**

Customer Table | Product_Dim | Store_Dim | Sales_Fact

PivotTable Fields
Choose fields to add to report:
Search
Count transby pyt
High or low value
Extract month and year
Product_Name
Category
More Tables...
Drag fields between areas below:
Filters
Columns
Rows
Values
Category
Sum of Total_Amount
Defer Layout Update
Update

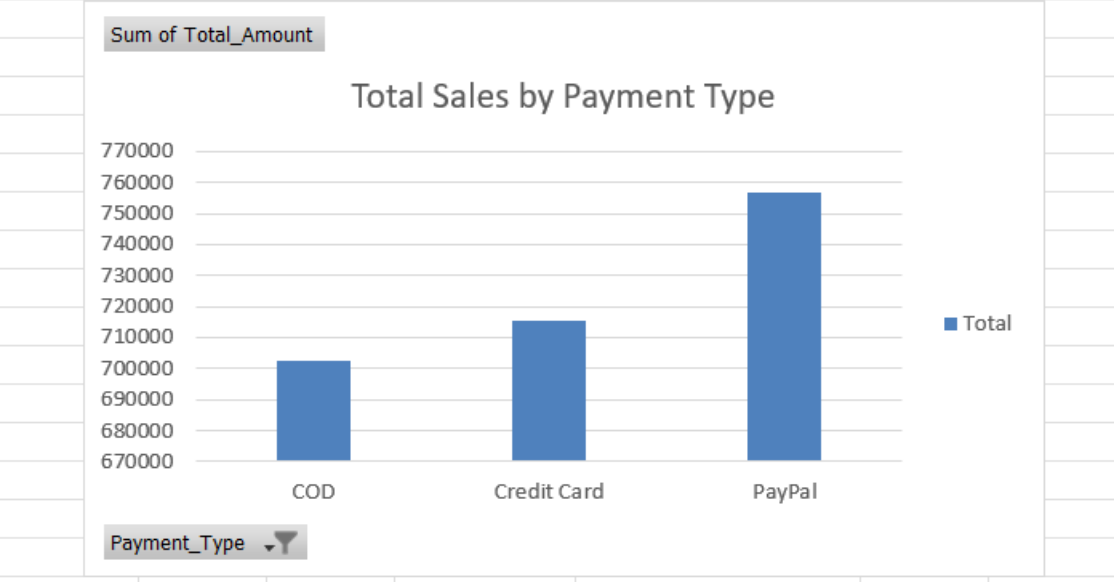
3. Descriptive statistics (mean, median, mode, standard deviation, etc.) using formulas or Analysis ToolPak

Statistic	Total Amount	Formula
Mean	1087.20	=AVERAGE(Sales_Fact!J2:J2001)
Median	868.72	=MEDIAN(Sales_Fact!J2:J2001)
Mode	No repeated value	=IFERROR(MODE.SNGL(Sales_Fact!J2),"No repeated value")
dard Devia	819.6189642	=STDEV.S(Sales_Fact!J2:J2001)
Minimum	18.46	=MIN(Sales_Fact!J2:J2001)
Maximum	3908.24	=MAX(Sales_Fact!J2:J2001)
Count	2000	=COUNT(Sales_Fact!J2:J2001)

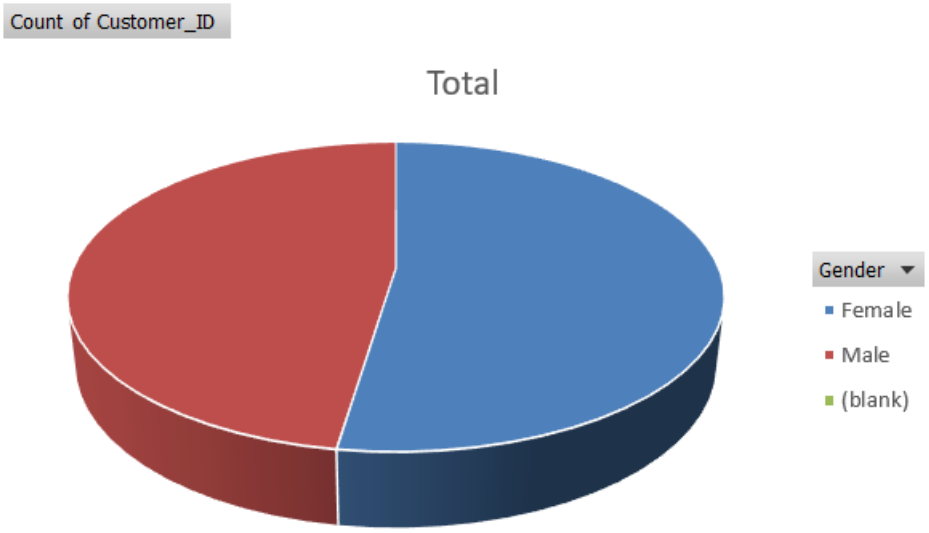
Customer Table | Product_Dim | Store_Dim | Sales_Fact | Descriptive StatisticsStatistic

4. Data visualization: create at least three chart types, build an interactive dashboard, and use basic PivotCharts

Row Labels	Sum of Total_Amount
COD	702459.4157
Credit Card	715180.711
PayPal	756754.5716
Grand Total	**2174394.698**



Row Labels	Count of Customer_ID
Female	262
Male	238
(blank)	
Grand Total	**500**



Row Labels	Count of Product_ID
Beauty	15
Clothing	18
Electronics	21
Home	21
Sports	25
Grand Total	**100**

